



Matteo Martinelli

Management Engineer

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Available to work abroad

 Website: <https://matteo-martinelli.github.io/> 



Professional Experience

- Nov 2024 – Now: Postdoctoral Research Associate – Research group of Pervasive Computing, **UNIMORE**

I participated in the **development of the Digital Twin technology** and the **analysis of its potential industrial applications**, also **contributing to the generation of leads** aimed at launching industrial research projects.

- Research topics:
 - Physical and digital interoperability, heterogeneity management, hierarchical architectures, Cyber-Physical Systems augmentation, twins orchestration, AI relationships;
 - Digital Twins and industrial applications, patterns of specialization, use cases investigation, AI pipelines and applications, synergies with constrained industrial devices, Digital Twin design techniques for industrial systems.
- Related activities
 - Research group activities **dissemination** and network construction in the **research and business domain**;
 - Industrial projects designs support for industrial research projects;
 - Coordination of thesis students in research and experimentation activities;
 - Organization of the IDETwin 2026 Workshop in Las Vegas, dedicated to advancing research and innovation in Industrial Digital Twin and Cyber-Physical Systems Intelligence .

- Nov 2024 – Now: **Freelance professional trainer**

- Teaching topics: Programming logics and design tools, Python, Java and OOP programming, data analysis with Python, Docker and containerization, principles of cloud computing, Artificial Intelligence and Machine Learning principles, Git and GitHub usage.

- Programs:
 - IFTS Software programming with AI tools - 2026 edition, organized by IFOA
 - MECFUTURE-TRANSITION_PRO - 2025 edition: "Information Systems Innovation: IoT, Machine Learning, and Artificial Intelligence", organized by W-Training
 - Full-Stack-Dev – Application Design and Development Technician - 2025 edition, organized by IFOA;
 - ITS DevSecOps - 2024/2025 and 2025/2026 editions, organized by Fondazione Olivetti and promoted by IFOA;
 - Women and Digital – 2024 edition, organized by IFOA;
 - ICT Skills for Employment, Inclusion and Sustainability – 2024 edition, organized by IFOA.

- Sept – Dec 2024: holder of the course "Data Science and Management", contract professor at the **University of Modena and Reggio Emilia, Italy**

- Teaching topics:
 - Python and OOP recall, data manipulation, preparation and visualization for ML with Numpy, Pandas, and Matplotlib;
 - Big data concepts, from traditional to NoSQL DBs, distributed file systems principles and computation, ERP systems;

I authorize the use of my personal data in compliance with the GDPR 679/16

- Provision of evaluation examinations.
 - Dec 2020 – Nov 2024: Research Grant – Research group of Pervasive Computing, UNIMORE
 - Research topics:
 - Application of eXplainable AI (XAI) in industrial contexts, causality for industrial root cause analysis, testing and simulations, autonomous development of smart agents;
 - Study of the role of Digital Twins for industrial contexts, synergies between AI and Digital Twins, patterns of interactions in Cyber Physical Production Systems (CPPS).
 - Related activities:
 - Research group activities dissemination in the research community at Italian and international scientific conferences;
 - Nov 2019 – Dec 2020: Focus Improvement & Safety Specialist, Full time – Ognibene Power
 - Main activities: I contributed to the **monitoring and improvement of industrial processes** in assigned areas, through planned projects and emerging departmental activities. I **successfully completed the 2020 audit**.
 - Jun – Nov 2019: Workplace Organization Support, Part time – Ognibene Power
 - Main activities: I contributed to the analysis and improvement of the layout of the assigned area. I also contributed to the time and method analysis activities and the updating of line setup information.
 - Jan – May 2019: Master's Degree Internship, Full time – Dana Motion System
 - Main activities: I **designed the assembly line** of a mechanical family of products with the goal of **achieving World Class performance**. Contextually, I **developed an AI-based Poka-Yoke Proof of Concept (POC)** to avoid frequently assembly errors. The result has been published on a scientific journal.
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Education & Training

Dec 2020 – May 2024: Ph.D. in Industrial Innovation Engineering – Research group of Pervasive Computing, UNIMORE

Thesis: Smart Digital Twins in Industry 4.0.

Oct 2022 – Apr 2023: Guest Researcher at ISW - *Institut für Steuerungstechnik der Werkzeugmaschinen* (Institute for Technologies of machine tools control) 

Topics: **Digital Twins for Industry 4.0 and Composability** - Universität Stuttgart, Stuttgart, Germany.

Jun 2022: ACAI TAILOR Summer School 2022 – *Prize for the best PhD project presentation* 

Topics: **Explainable and Trustworthy AI** - Universitat Politècnica de Catalunya, Barcelona, Spain.

Oct 2019: Management Engineering Master's Degree, UNIMORE 105/110

Thesis: "An assembly line re-engineering: The Dana Motion System case".

Jul 2017: participation at the Erasmus+ Program – Krakow, Poland 

Dec 2015: Management Engineering Bachelor's Degree, UNIMORE 95/110

Thesis: "Implementation of a reading and translation system of DXF files into two axis PLC instructions".

Technologies applied: PLC, cartesian robot, C#, code design, design patterns.

Computer Skills:

- Programming Languages: C#, **Java**, **Python**, **JavaScript**.
- Web technologies: **HTML**, **CSS**.
- Programming Libraries: Keras, Numpy, Pandas, Scikit-Learn, PuLP.
- DB Technologies and developing tools: SQL, document-oriented DBs, Docker, Git, UML.
- Operating Systems: Windows, Ubuntu, MacOS.
- Software: Anylogic – Material Handling and Process Model libraries, Ms Office suite.

Languages

English (C1 grade);

German (A2 grade);

Italian (mother language).

Soft Skills

- Precise
 - Reliable
 - Communicative
 - Loyal
 - Open minded, *learning* and *people* oriented
 - Assertive
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Other Information

- Attachment 1: Certificates and Badges.pdf

Date

Matteo Martinelli

15/01/2026



Certificates and Badges



Software and technologies for Data Science

- Date: March 2021;
- Provided by: the Big DataLab project sponsored by the ESF.

https://openbadges.bestr.it/public/assertions/rh6U8Z45S8GyXH_Wx6koSg



Internet of Things and Predictive Analysis

- Date: June 2021;
- Provided by: the Big DataLab project sponsored by the ESF.

<https://openbadges.bestr.it/public/assertions/JNZKBoomR8GTvl-0YMc3lQ>



Deep Learning Applications

- Date: June 2021;
- Provided by: the Big DataLab project sponsored by the ESF.

<https://openbadges.bestr.it/public/assertions/bVoUXhQHRG2vhfunvjMTCw>

Date

15/01/2026

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